

SF1811 Optimization

Lecture 4: Network flow and transportation problems

Chapter 7

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Network flow and transportation problems

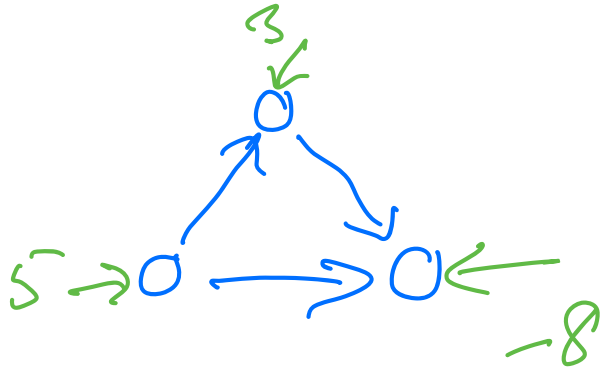
Main question:

How can the special structure of a flow problem simplify the simplex method?

Today:

- Minimum-cost flow.
- Incidence matrices and flow balance.
- Bases and spanning trees.
- Simplex iterations as cycle adjustments.
- The transportation problem.

Minimum-cost flow



A network consists of

- a finite set of nodes $\mathcal{N}$;
- a set of directed edges $\mathcal{E}$.

For every edge $(i, j) \in \mathcal{E}$:

$$x_{ij} = \text{flow on edge } (i, j), \qquad c_{ij} = \text{cost per unit flow.}$$

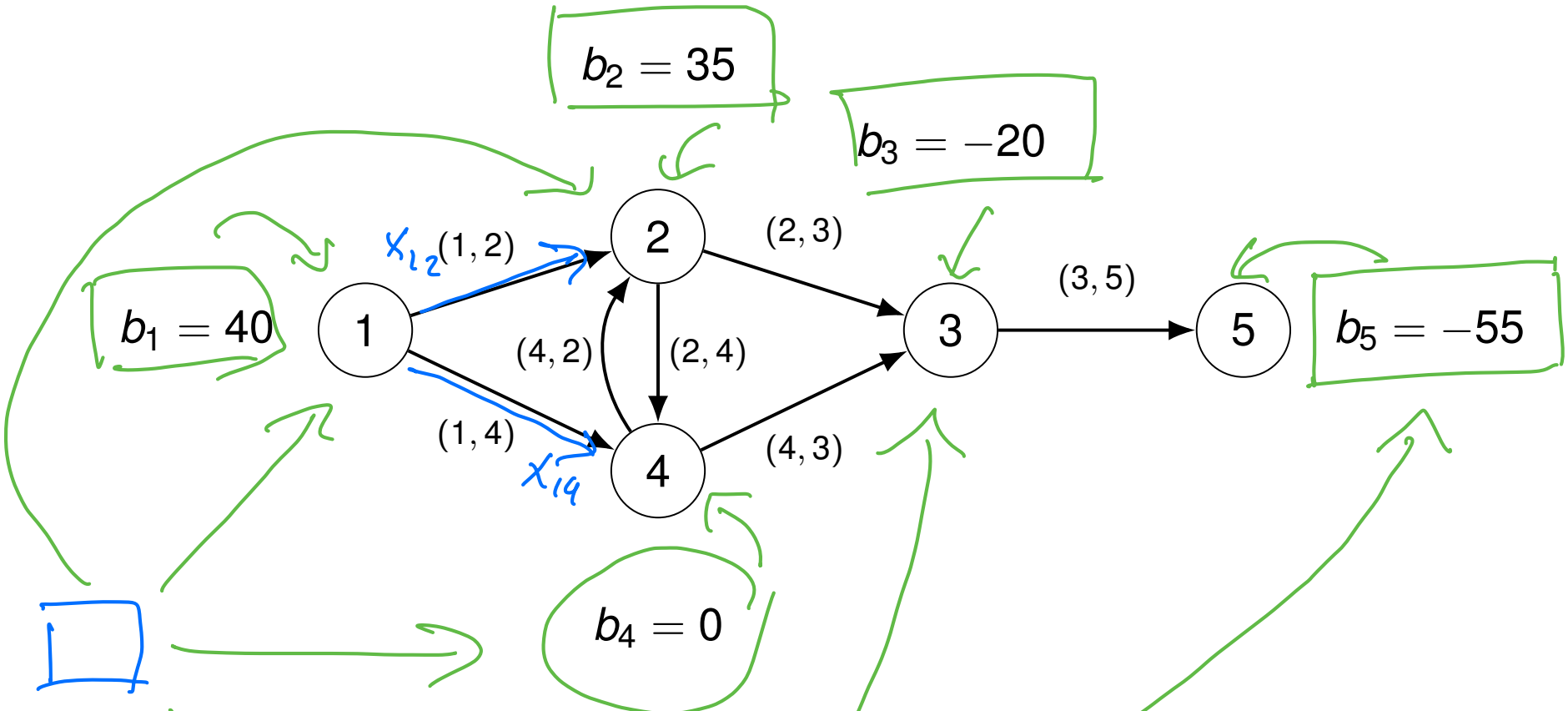
At node i:

$$b_i = \begin{cases} > 0, & \text{external supply,} \\ < 0, & \text{external demand,} \\ = 0, & \text{intermediate node.} \end{cases}$$

We assume that total supply equals total demand:

$$\sum_{i \in \mathcal{N}} b_i = 0.$$

A network flow example



The edge order is

$$(1,2), (1,4), (2,4), (4,2), (2,3), (4,3), (3,5).$$

	(1,2)	(1,4)	(2,4)	(4,2)	(2,3)	(4,3)	(3,5)	
	1	1						40
Flow balance	-1		1	-1	1			35
					-1	-1	1	-20
at every node		-1	-1	1		1		0
							-1	-55

Flow balance and the incidence matrix

For each node:

$$\text{outflow} - \text{inflow} = b_i.$$

The node-edge incidence matrix $\widetilde{A}$ is defined by

$$\widetilde{a}_{ik} = \begin{cases} 1, & \text{if edge } k \text{ starts at node } i, \\ -1, & \text{if edge } k \text{ ends at node } i, \\ 0, & \text{otherwise.} \end{cases}$$

The minimum-cost flow problem is

$$\begin{array}{ll} \text{minimize} & c^{\mathsf{T}} x \\ \text{subject to} & \widetilde{A}x = \widetilde{b}, \\ & x \geq 0. \end{array}$$

Each column of $\widetilde{A}$ contains one 1 and one -1.

The incidence matrix for the example

The edge order is $(1,2)\ (1,4)\ (2,4)\ (4,2)\ (2,3)\ (4,3)\ (3,5)$

$$c^T = \begin{pmatrix} 2 & 5 & 2 & 2 & 1 & 1 & 2 \end{pmatrix}$$

$$\widetilde{A} = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \\ 0 & -1 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}, \quad \widetilde{b} = \begin{pmatrix} 40 \\ 35 \\ -20 \\ 0 \\ 55 \end{pmatrix}$$

$$x = \begin{pmatrix} x_{12} \\ x_{14} \\ x_{24} \\ x_{42} \\ x_{23} \\ x_{43} \\ x_{35} \end{pmatrix}$$

Each column contains one 1 and one -1.

One redundant flow-balance equation

Since every column of $\widetilde{A}$ contains one 1 and one -1,

$$e^{\mathsf{T}}\widetilde{A} = 0, \quad \text{with} \quad e^{T} = \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix}.$$

Thus, the rows of $\widetilde{A}$ are linearly dependent.

Since $e^{\mathsf{T}}\widetilde{b} = \sum_{i \in \mathcal{N}} b_i = 0$, one node-balance equation follows from all the others.

Delete one row and the corresponding component of $\widetilde{b}$:

$$\begin{array}{ll} \text{minimize} & c^{\mathsf{T}}x \\ \text{subject to} & Ax = b, \\ & x \geq 0, \end{array}$$

where $A \in \mathbb{R}^{(m-1)\times n}$ has full row rank for a connected network.

We will see constructively

Every basis therefore contains $m - 1$ edges.

Spanning trees and bases

A set $\mathcal{T}$ of edges is a spanning tree if

- every node is included;
- the edges connect the network;
- the edges contain no cycle when directions are ignored.

Theorem

A set of $m-1$ columns of A is linearly independent if and only if the corresponding $m-1$ edges form a spanning tree.

For a network flow problem, choosing a basis means choosing a spanning tree.

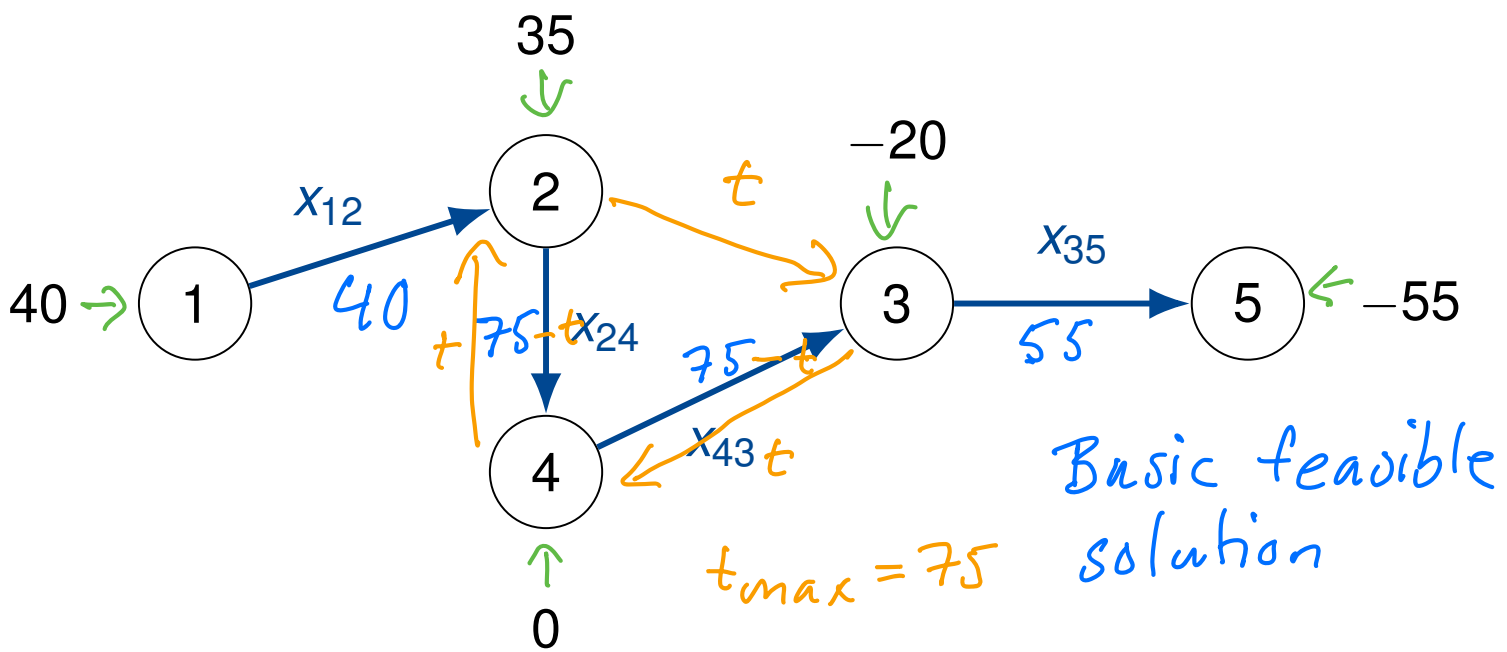
A spanning-tree basis

Choose

$$\mathcal{T} = \{(1,2),(2,4),(4,3),(3,5)\}.$$

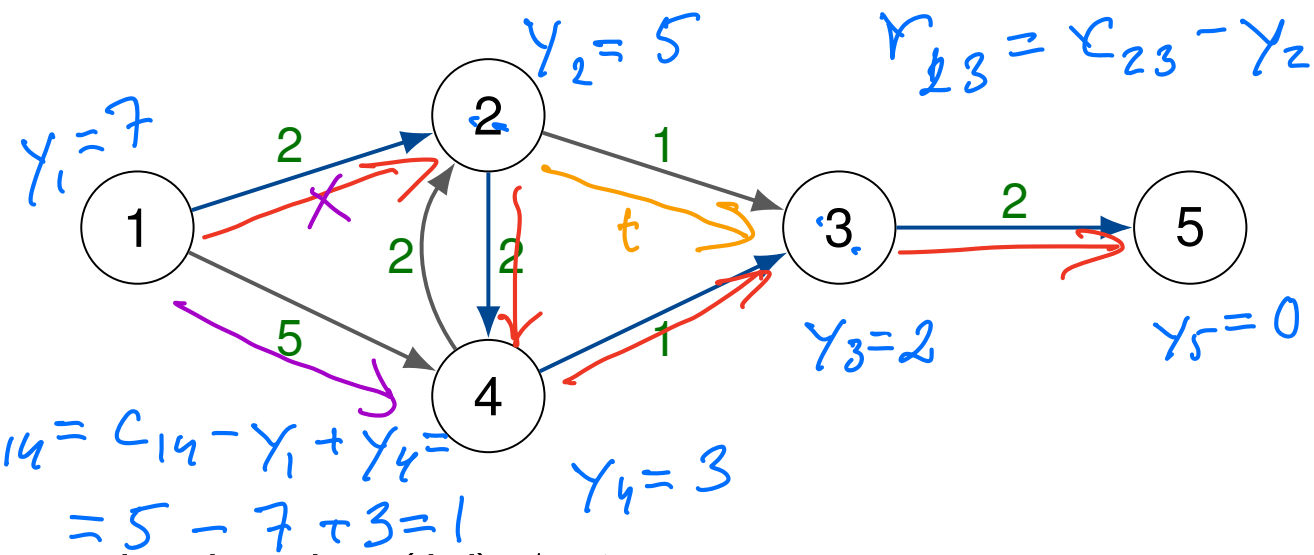
The nonbasic flows are

$$x_{14} = x_{42} = x_{23} = 0.$$



Node potentials and reduced costs

- For the tree basis, the simplex multipliers satisfy $A_B^T y = c_B$.
- Introduce one potential y_i for each node and set $y_5 = 0$.
- For every basic edge $(i,j) \in \mathcal{T}$, $y_i - y_j = c_{ij}$.

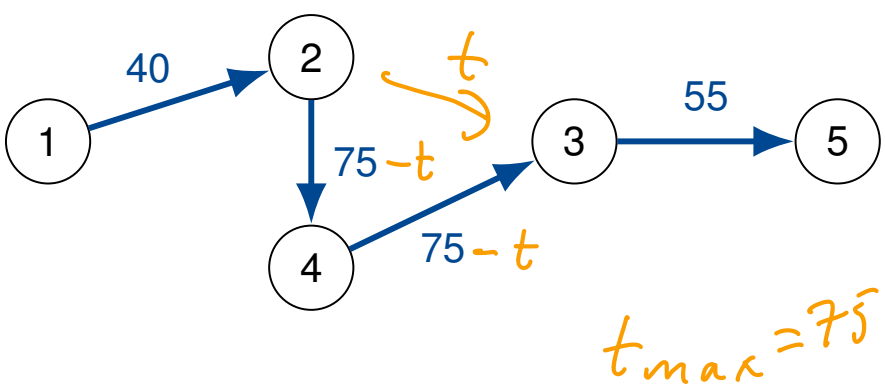


- For every nonbasic edge $(i,j) \notin \mathcal{T}$, $r_{ij} = c_{ij} - y_i + y_j$.
- If $r_{ij} < 0$, increasing x_{ij} improves the objective value.

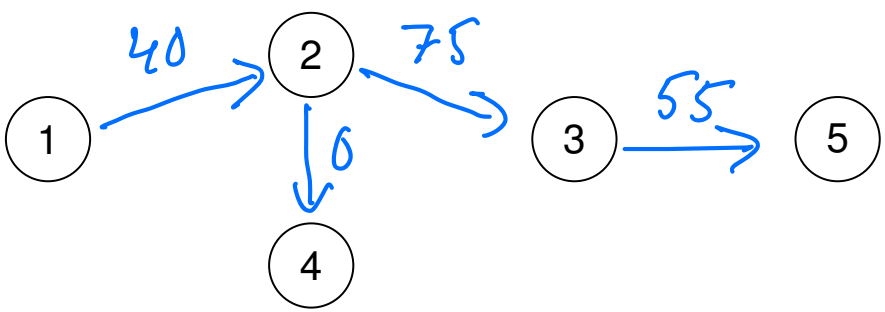
Finding a new spanning tree

- Add the entering edge (i, j) to the spanning tree.
- The entering edge and the tree path between its endpoints form a unique cycle. Only flows on this cycle change.
- Determine the changes in the basic flows when $x_{ij} = t$.
- Determine $t_{\max}$ and the resulting new spanning tree.

Determine $t_{\max}$

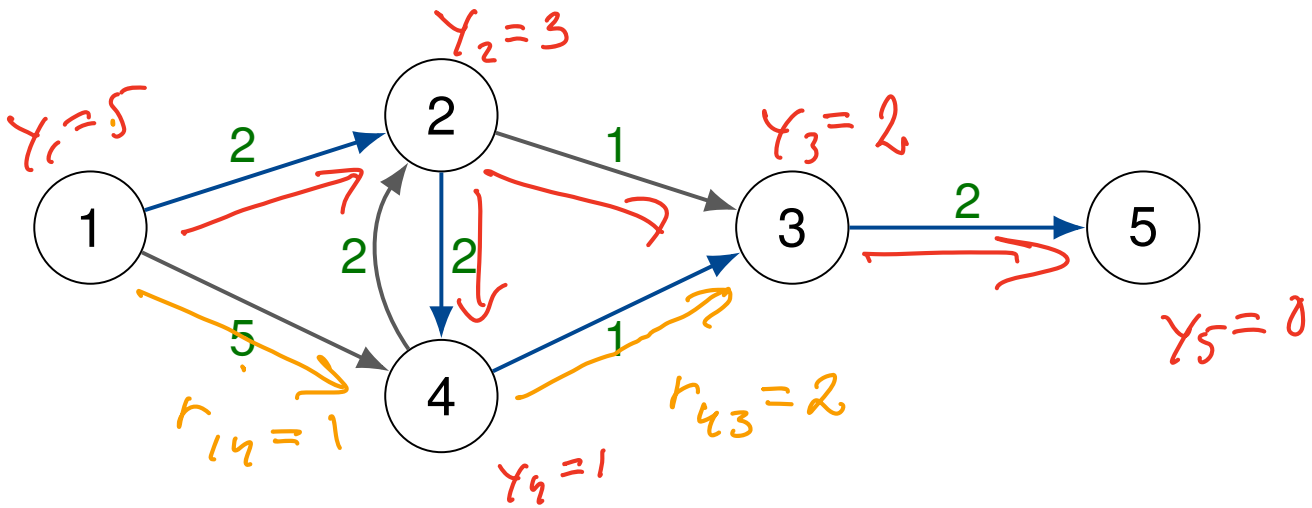


New basic feasible solution



Node potentials and reduced costs

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- For every basic edge $(i,j) \in \mathcal{T}$, $y_i - y_j = c_{ij}$.



- For every nonbasic edge $(i,j) \notin \mathcal{T}$, $r_{ij} = c_{ij} - y_i + y_j$.
- If $r_{ij} < 0$, increasing x_{ij} improves the objective value.

$r_{ij} \geq 0 \Rightarrow$ optimal

Network simplex interpretation

For a minimum-cost flow problem:

1. A basis is represented by a spanning tree.
2. Basic flows are determined from node balance on the tree.
3. Simplex multipliers are represented by node potentials.
4. Reduced costs are computed edge by edge:

$$r_{ij} = c_{ij} - y_i + y_j.$$

5. An entering edge creates one cycle.
6. Flow is adjusted around the cycle until a basic edge reaches zero.

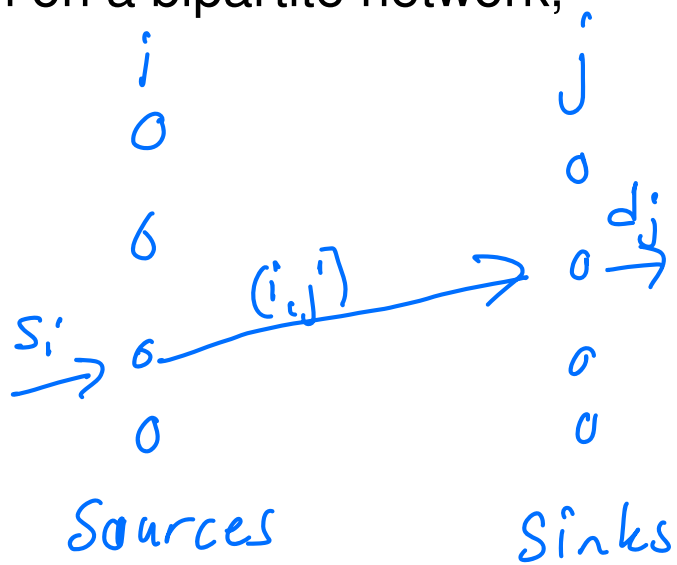
The tree and the cycle replace much of the matrix algebra.

The transportation problem

The transportation problem is a minimum-cost flow problem on a bipartite network, with one node for each source and destination.

There are ℓ sources and k destinations.

$s_i =$ supply at source i,
$d_j =$ demand at destination j,
$c_{ij} =$ unit transportation cost,
$x_{ij} =$ amount shipped from i to j.



The balanced transportation problem is

$$\begin{aligned}
\text{minimize} \quad & \textstyle\sum_{i=1}^{\ell} \sum_{j=1}^{k} c_{ij} x_{ij} \\
\text{subject to} \quad & \textstyle\sum_{j=1}^{k} x_{ij} = s_i, \quad i = 1, \ldots, \ell, \\
& \textstyle -\sum_{i=1}^{\ell} x_{ij} = -d_j, \quad j = 1, \ldots, k, \\
& x_{ij} \geq 0.
\end{aligned}$$

Assumption

$$\sum_{i=1}^{k} s_i = \sum_{j=1}^{l} d_j$$

A transportation example

c_{ij}	$j=1$	$j=2$	$j=3$	s_i
$i=1$	4	5	5	55
$i=2$	4	3	4	50
d_j	40	25	40	

The problem is balanced:

$$55 + 50 = 40 + 25 + 40 = 105.$$

A basic solution has

$$\ell + k - 1 = 2 + 3 - 1 = 4$$

basic variables.

How can we construct an initial basic feasible solution directly in the table?

The northwest corner rule

Start in the northwest corner and allocate as much as possible.

	$j=1$	$j=2$	$j=3$	s_i
$i=1$	40	15		55
$i=2$		10	40	50
d_j	40	25	40	

$\Rightarrow$ Basic feasible solution.

1. Allocate $\min\{s_i, d_j\}$ in the current cell.
2. Reduce the remaining supply and demand.
3. Move east if the demand is exhausted, and south if the supply is exhausted.
4. Continue to the southeast corner.

Transportation simplex: potentials and reduced costs

- Introduce node potentials u_i and v_j and set $v_3 = 0$.
- For every basic variable x_{ij},

$$u_i - v_j = c_{ij}.$$

- For every nonbasic variable x_{ij},

$$r_{ij} = c_{ij} - u_i + v_j.$$

u_i source nodes

v_j sink nodes

Basic var $c_{ij} - u_i + v_j = 0$

Node potentials

c_{ij}	$j=1$	$j=2$	$j=3$	u_i
$i=1$	4	5	5	6
$i=2$	4	3	4	4
v_j	2	1	0	

Reduced costs

r_{ij}	$j=1$	$j=2$	$j=3$
$i=1$	–	–	-1
$i=2$	2	–	–

$r_{13} = -1 < 0$

Increase x_{13} from 0.

The potentials are the simplex multipliers in network notation.

Transportation simplex: the cycle adjustment

- Set the entering variable $x_{ij} = t$.
- Keeping all row and column sums fixed gives a unique change in the basic variables.
- Determine $t_{\max}$ and the resulting new basic feasible solution.

Determine $t_{\max}$

x_{ij}	$j=1$	$j=2$	$j=3$	s_i
$i=1$	40	$15-t$	t	55
$i=2$		$10+t$	$40-t$	50
d_j	40	25	40	

$t_{max} = 15$

New basic feasible solution

x_{ij}	$j=1$	$j=2$	$j=3$	s_i
$i=1$	40		15	55
$i=2$		25	25	50
d_j	40	25	40	

⇒ optimal solution

Transportation simplex: potentials and reduced costs

- Introduce node potentials u_i and v_j and set $v_3 = 0$.
- For every basic variable x_{ij},

$$u_i - v_j = c_{ij}.$$

- For every nonbasic variable x_{ij},

$$r_{ij} = c_{ij} - u_i + v_j.$$

Node potentials

c_{ij}	$j=1$	$j=2$	$j=3$	u_i
$i=1$	4	5	5	5
$i=2$	4	3	4	4
v_j	1	1	0	

Reduced costs

r_{ij}	$j=1$	$j=2$	$j=3$
$i=1$	–	1	–
$i=2$	1	–	–

$r_{ij} \geq 0 \Rightarrow$ optimal

The potentials are the simplex multipliers in network notation.

Why the special structure matters

For network flow and transportation problems:

- bases are represented by trees;
- an entering variable creates a unique cycle;
- directions are found by alternating changes around the cycle;
- node potentials give the simplex multipliers;
- reduced costs are calculated locally on edges;
- integer data lead to integer basic feasible solutions.

The general simplex method remains unchanged, but the network structure makes the calculations transparent.

Main ideas

After this lecture, you should be able to:

- formulate a minimum-cost flow problem;
- construct and interpret a node-edge incidence matrix;
- explain why one flow-balance equation is redundant;
- relate a network-flow basis to a spanning tree;
- compute node potentials and reduced costs;
- interpret a simplex direction as a cycle adjustment;
- formulate and solve a small transportation problem.

In structured LP problems, graphical structure can replace much of the matrix algebra.